PHYS4055

Atomic physics and quantum optics

Week 7

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(from next week: Jacinda Ginges)

Atomic physics overview

- **Week 7**

Hydrogen, hydrogen-like ions, and electrons in spherical potentials

- **Week 8**

Shell filling, electron coupling, and angular momenta

- **Week 9**

Angular momentum exploration, self-consistent fields, Dirac equation

- **Week 10**

Interaction with external fields

Seminars begin

- **Week 11**

Your seminars!

- **Week 12**

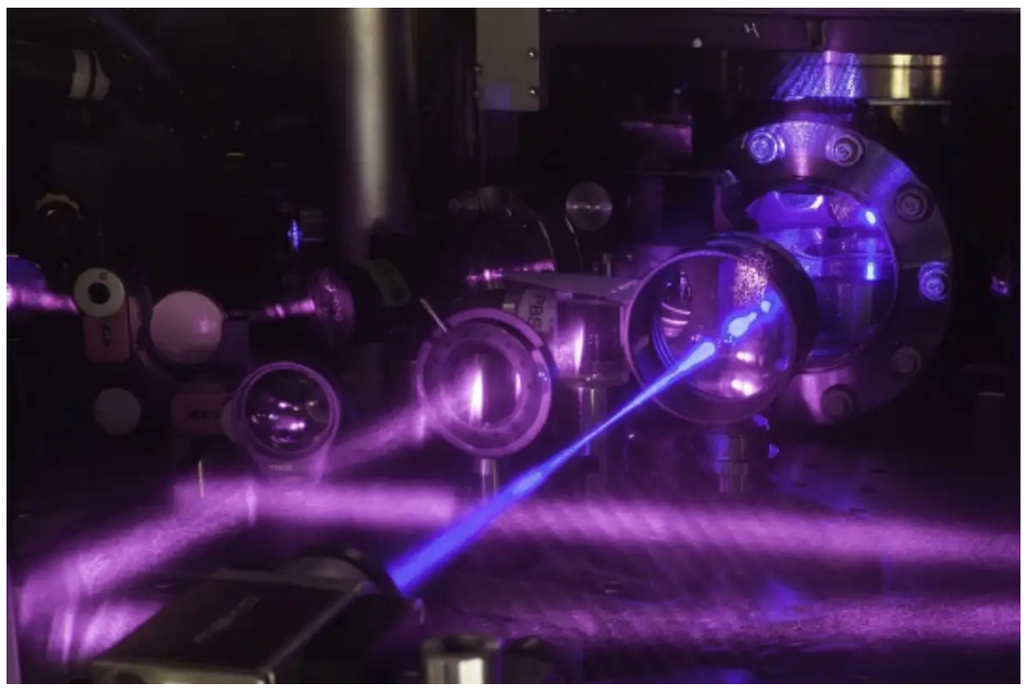
Atom cooling and trapping

Clocks and isotope shift

- **Week 13**

Hot topics

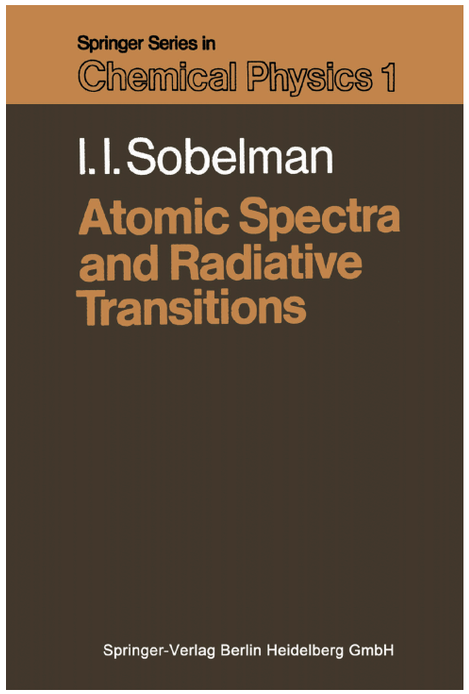
Searches for new physics in precision atomic experiments



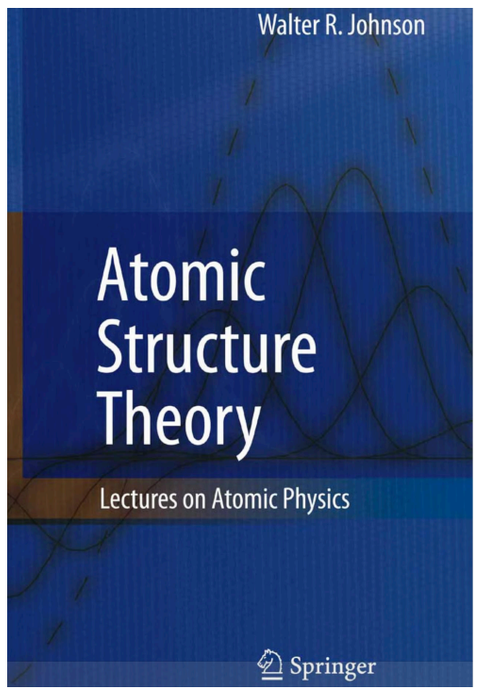
JILA Sr clock

Atomic physics resources

Primary texts:

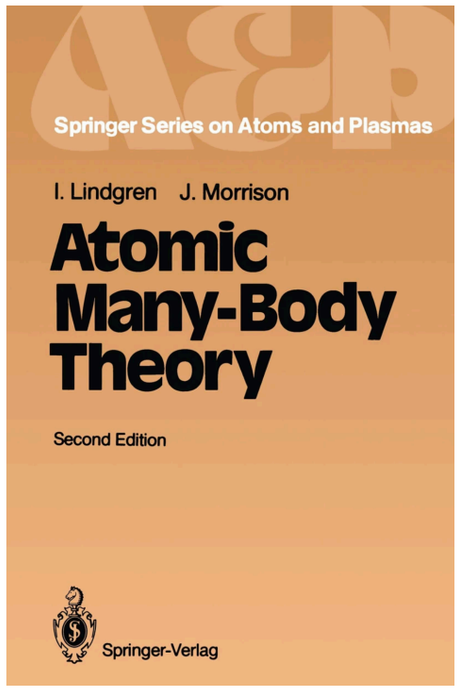


Full e-text through library and Blackboard

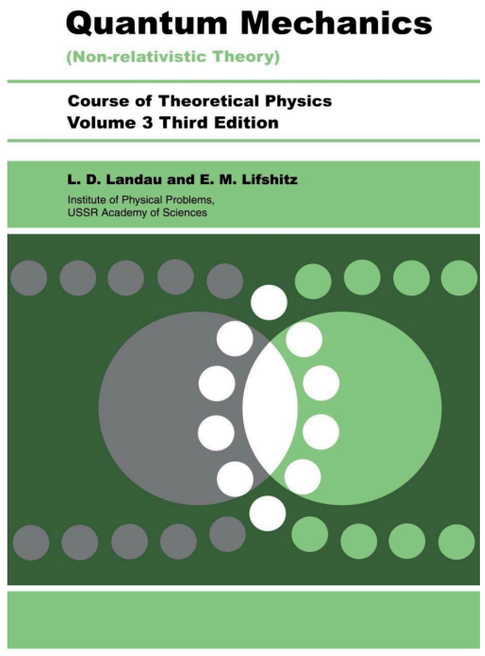


Full document in Blackboard

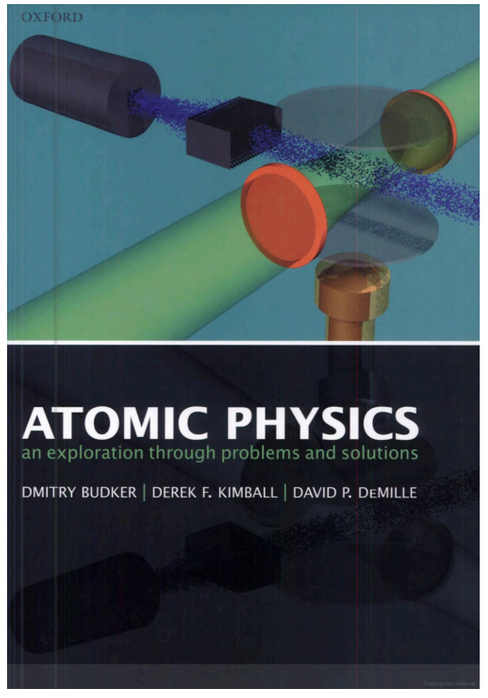
Other useful texts:



Full e-text through library



Full e-text through library



Selected pages in Blackboard

Staying on top of content; support

Readings: specified by week for the atomic physics section in Blackboard

Mathematica Notebooks: for use in applied classes, problem sets, and to support and enhance understanding and learning

— 3 Notebooks spanning aspects of the atomic physics content

Homework: install Mathematica on your laptop, and bring with you to the applied classes

This week, we will:

- Introduce the spectra and wave functions of H and H-like systems, and consider:
  - Relativistic effects
  - Fine structure
  - Lamb shift
  - Hyperfine structure

Hydrogen and hydrogen-like ions

Hydrogen: electron with mass *m* and electric charge -|*e*| in field of proton with mass *M>>m* and electric charge |*e*|.

Reduced mass: $\mu = \dfrac{mM}{m+M} \simeq m \qquad (M \gg m)$

Hydrogen-like ion: single electron in the field of a nucleus with Z protons and N neutrons. Nuclear electric charge: Z|e|.

Take proton/nucleus at origin, and consider proton charge distribution to be point-like.

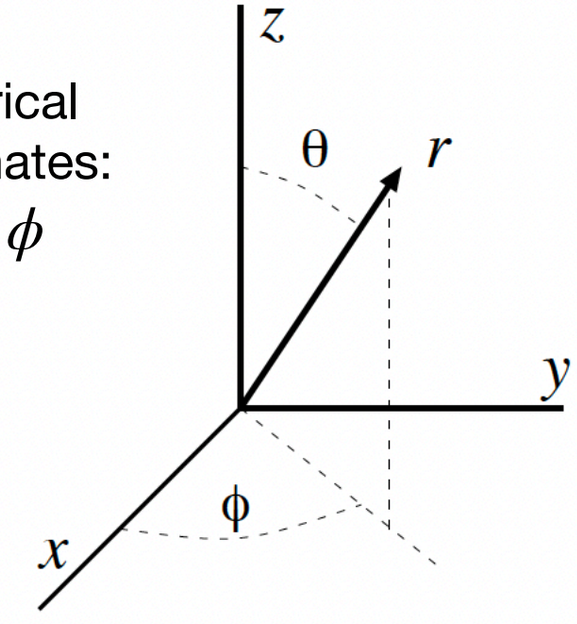
$\Rightarrow$ (Coulomb) potential energy $V_{\text{nuc}}(r) = -\dfrac{Ze^2}{4\pi\epsilon r}$

Schrödinger equation: $\left(\dfrac{p^2}{2m} + V(r)\right)\psi = \varepsilon\psi$

$$\Rightarrow \left(-\frac{\hbar^2\nabla^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r}\right)\psi = \varepsilon\psi$$

Has solutions: $\psi(\mathbf{r}) = \dfrac{1}{r}P(r)Y(\theta,\phi)$

Spherical coordinates: r, θ, ϕ



$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

Atomic units:

$$|e| = m = \hbar = 4\pi\epsilon_0 = 1$$

Hydrogen and hydrogen-like ions

Obtain two equations: one radial and one angular

Radial: $$\frac{d^2P}{dr^2} + \frac{2m}{\hbar^2}\left(E - V(r) - \frac{\lambda\hbar^2}{2mr^2}\right)P = 0$$

Angular: $$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2} + \lambda Y = 0$$

A separation constant λ was introduced. If $\lambda = l(l+1)$, then

$$Y(\theta,\phi) = Y_{lm}(\theta,\phi)$$ spherical harmonics.

Here, *l* is an integer, *l=0,1,2,… .*

H and H-like wave functions have the form: $$\psi_{nlm}(\mathbf{r}) = \frac{1}{r}P_{n,l}(r)Y_{l,m}(\theta,\phi)$$

Normalisation of wave function $$\int \psi^\dagger(\mathbf{r})\psi(\mathbf{r})d^3r = 1$$

and expectation value of any operator: $$\langle\hat{O}\rangle = \int \psi^\dagger(\mathbf{r})\hat{O}\psi(\mathbf{r})d^3r$$

Hydrogen and hydrogen-like ions

Exercise. Derive the two separated angular and radial equations in the previous slide (*homework*).

1. Substitute $\psi(\mathbf{r}) = P(r)Y(\theta,\phi)/r$ into the Schrödinger equation.
2. Use $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$

   and $\frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{d}{dr}\left(\frac{P}{r}\right)\right] = \frac{1}{r}\frac{d^2P}{dr^2}$.
3. Divide by $P(r)Y(\theta,\phi)/r$.
4. Collect all radial terms on one side and angular terms on the other.
5. Set both sides equal to the separation constant λ.

Exercise. Find $\langle L^2\rangle$ and $\langle L_z\rangle$.

$$L^2Y_{lm} = l(l+1)\hbar^2Y_{lm}\,, \quad L_zY_{lm} = m\hbar Y_{lm} \quad \Rightarrow \quad \langle L^2\rangle = l(l+1)\hbar^2, \quad \langle L_z\rangle = m\hbar$$

Hydrogen and hydrogen-like ions

Exercise. Explore conditions of normalisation on Y and P.

We start from $\int \psi^*_{nlm}(\mathbf{r})\psi_{nlm}(\mathbf{r})\, d^3r = 1$, with

$$\psi_{nlm}(\mathbf{r}) = \frac{P_{nl}(r)}{r} Y_{lm}(\theta,\phi)\,, \quad d^3r = r^2\, dr\, d\Omega\,, \quad d\Omega = \sin\theta\, d\theta\, d\phi$$

Substituting: $$\int_0^\infty \frac{P^*_{nl}(r)}{r}\frac{P_{nl}(r)}{r} r^2\, dr \int Y^*_{lm}(\theta,\phi) Y_{lm}(\theta,\phi)\, d\Omega = 1$$

$$\Rightarrow \left[\int_0^\infty |P_{nl}(r)|^2\, dr\right]\left[\int |Y_{lm}(\theta,\phi)|^2\, d\Omega\right] = 1$$

$$\Rightarrow \int_0^\infty |P_{nl}(r)|^2\, dr = 1\,, \quad \int_0^{2\pi} d\phi \int_0^\pi |Y_{lm}(\theta,\phi)|^2 \sin\theta\, d\theta = 1$$

Functions with different quantum numbers are orthonormal:

$$\int_0^\infty P^*_{nl}(r) P_{n'l'}(r)\, dr = \delta_{nn'}\delta_{ll'}\,, \quad \int Y^*_{lm} Y_{l'm'}\, d\Omega = \delta_{ll'}\delta_{mm'}$$

Hydrogen and hydrogen-like ions

Exact solution for the radial wave function,

$$P_{nl}(r) = N_{nl}(2Zr/n)^{l+1}e^{-Zr/n}F(-n+l+1,2l+2,2Zr/n)\,, \qquad N_{nl} = \frac{1}{n(2l+1)!}\sqrt{\frac{Z(n+l)!}{(n-l-1)!}}$$

N_{nl} is the normalisation constant, F is a hypergeometric function.

n is the principal quantum number, *n*=1,2,3,..

l is the orbital angular momentum quantum number, $0 \le l < n$

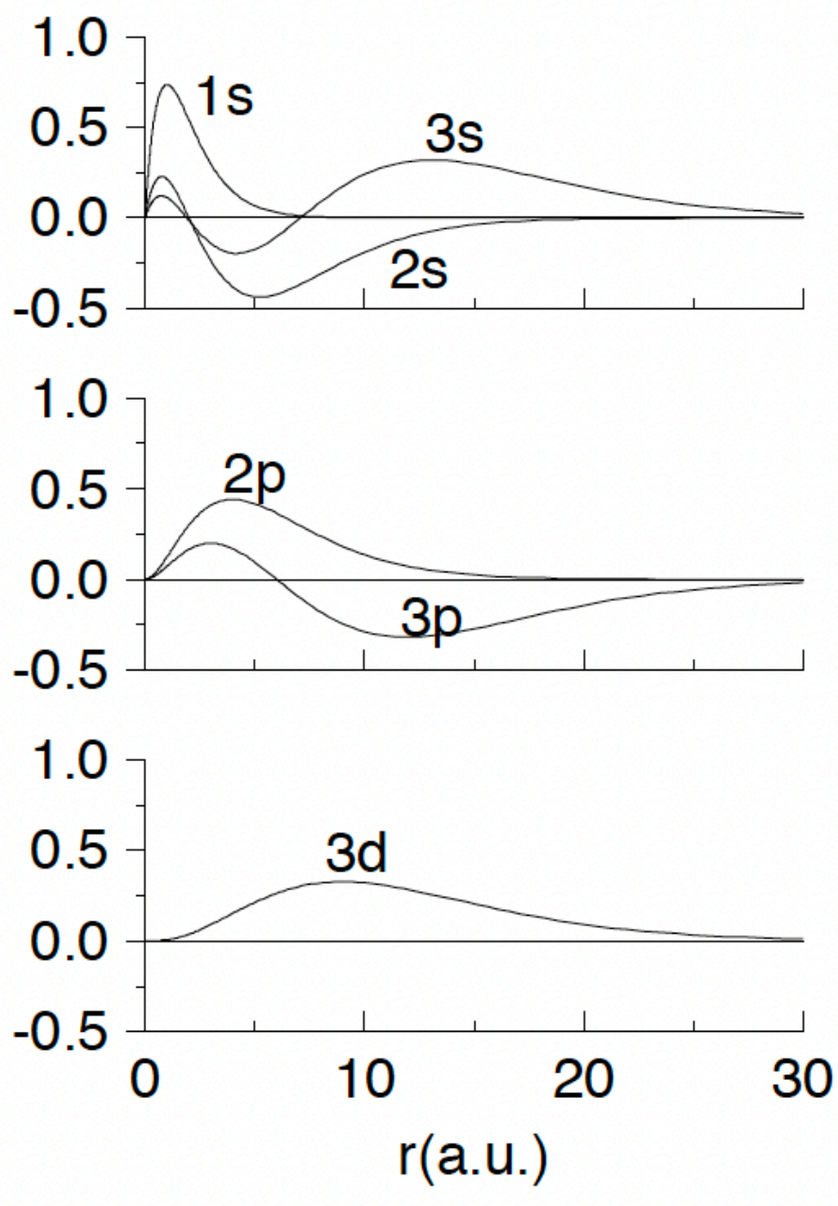
m is the magnetic quantum number (projection of *l* on the z-axis), $-l < m < l$

And $n = n_0 + l + 1$, where n_0 is the "number of zeros"

Note on spectroscopic notations: *l* = 0, 1, 2, 3,…
s, p, d, f,..

State with *n*=1, *l*=0: 1s
n=3, *l*=2: 3d

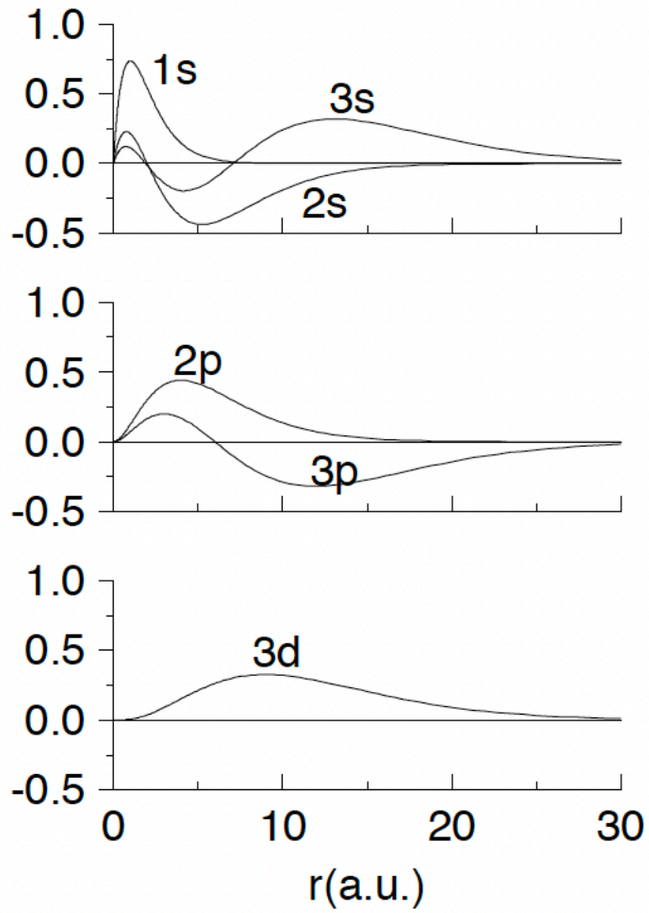
<u>Exercise</u>: check that $n = n_0 + l + 1$ by considering the wave functions to the right



First few functions

radial functions P_{nl} $\qquad 0 \le l < n$

$$
\begin{aligned}
P_{10}(r) &= 2Z^{3/2}\, re^{-Zr}\,, \\
P_{20}(r) &= \frac{1}{\sqrt{2}} Z^{3/2}\, re^{-Zr/2}\left(1-\frac{1}{2}Zr\right)\,, \\
P_{21}(r) &= \frac{1}{2\sqrt{6}} Z^{5/2}\, r^2 e^{-Zr/2}\,, \\
P_{30}(r) &= \frac{2}{3\sqrt{3}} Z^{3/2}\, re^{-Zr/3}\left(1-\frac{2}{3}Zr+\frac{2}{27}Z^2r^2\right)\,, \\
P_{31}(r) &= \frac{8}{27\sqrt{6}} Z^{5/2}\, r^2 e^{-Zr/3}\left(1-\frac{1}{6}Zr\right)\,, \\
P_{32}(r) &= \frac{4}{81\sqrt{30}} Z^{7/2}\, r^3 e^{-Zr/3}.
\end{aligned}
$$



spherical harmonics Y_{lm}

$-l \le m \le l$

$$
\begin{aligned}
Y_{00} &= \sqrt{\tfrac{1}{4\pi}} \\
Y_{10} &= \sqrt{\tfrac{3}{4\pi}}\,\cos\theta & Y_{1,\pm1} &= \mp\sqrt{\tfrac{3}{8\pi}}\,\sin\theta\; e^{\pm i\phi} \\
Y_{20} &= \sqrt{\tfrac{5}{16\pi}}\,(3\cos^2\theta-1) & Y_{2,\pm1} &= \mp\sqrt{\tfrac{15}{8\pi}}\,\sin\theta\cos\theta\; e^{\pm i\phi} \\
Y_{2,\pm2} &= \sqrt{\tfrac{15}{32\pi}}\,\sin^2\theta\; e^{\pm 2i\phi} \\
Y_{30} &= \sqrt{\tfrac{7}{16\pi}}\,\cos\theta\,(5\cos^2\theta-3) & Y_{3,\pm1} &= \mp\sqrt{\tfrac{21}{64\pi}}\,\sin\theta\,(5\cos^2\theta-1)\; e^{\pm i\phi} \\
Y_{3,\pm2} &= \sqrt{\tfrac{105}{32\pi}}\,\cos\theta\sin^2\theta\; e^{\pm 2i\phi} & Y_{3,\pm3} &= \mp\sqrt{\tfrac{35}{64\pi}}\,\sin^3\theta\; e^{\pm 3i\phi}
\end{aligned}
$$

Energies (gross structure)

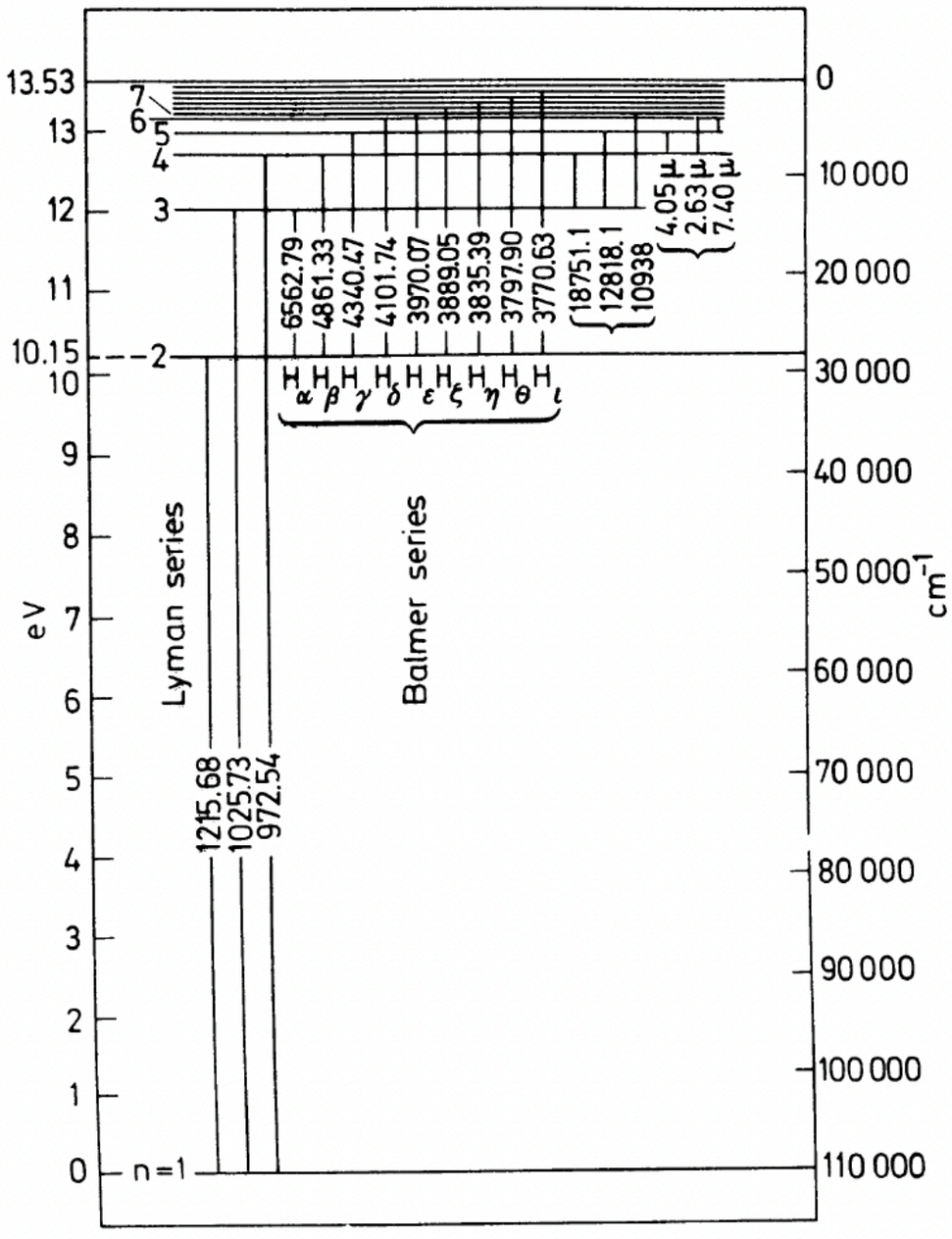
Energy solution:

$$E = E_n = -\frac{Z^2}{2n^2}\frac{me^4}{\hbar^2} = -\frac{Z^2}{2n^2}\text{a.u.} = -\frac{Z^2}{n^2}\text{Ry}$$

Ground state: $E_1 = -13.6\,Z^2\,\text{eV}$

Ground-state wave function:

$$\psi_{100} = P_{10}(r)/rY_{00}(\theta,\phi) = 2Z^{3/2}e^{-Zr}\sqrt{1/4\pi}$$



H spectrum

Exercise: Find the first excited state energy. How many states are there with this energy? Give their quantum numbers and write out their wave functions.

For hydrogen: $E_2 = -\frac{13.6}{4}\ \text{eV} = -3.4\ \text{eV}$

States with this energy:

$2s:\quad n=2,\ l=0,\ m=0$

$2p:\quad n=2,\ l=1,\ m=-1,0,+1$

Wavefunctions: $\psi_{200}(\mathbf{r}) = \frac{P_{20}(r)}{r}Y_{00}(\theta,\phi)\,,\qquad \psi_{21m}(\mathbf{r}) = \frac{P_{21}(r)}{r}Y_{1m}(\theta,\phi),\quad m=-1,0,+1$

Energies (gross structure)

Difference between two energy levels:

$$E_{n'} - E_n = -Z^2\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)\mathrm{Ry}$$

Photon emission/absorption: $h\nu = |E_{n'} - E_n|\,, \quad \lambda = \frac{c}{\nu}$

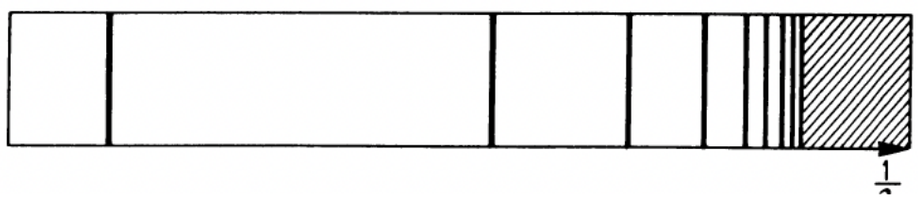
$n' = 1$: $\frac{1}{\lambda} = R\left(1 - \frac{1}{n^2}\right)$, $n = 2,3,4, \ldots$ Lyman series

$n' = 2$: $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$, $n = 3,4,5, \ldots$ Balmer series

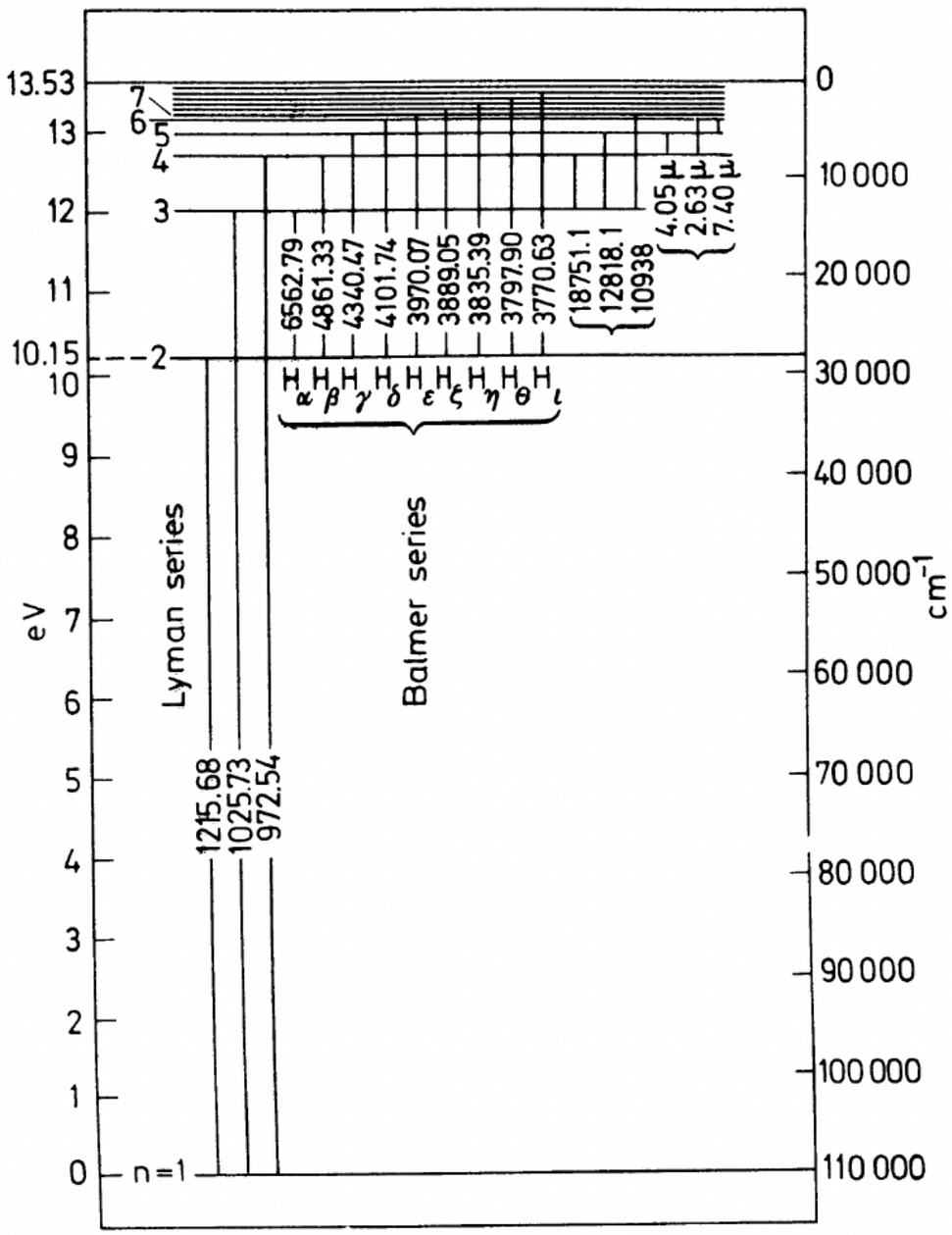
$n' = 3$: $\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$, $n = 4,5,6, \ldots$ Paschen series

$n' = 4$: $\frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n^2}\right)$, $n = 5,6,7, \ldots$ Brackett series

$n' = 5$: $\frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{n^2}\right)$, $n = 6,7,8, \ldots$ Pfund series



Emission spectrum for H



H spectrum

Exercises

Calculate expectation value of powers p of r, $\langle r^p \rangle_{nl}$, for $p = 1,2,-1,-2$.

$$\langle r^p \rangle_{nl} \equiv \langle \psi | \, r^p \, | \psi \rangle \equiv \int \psi^*_{nlm}(\mathbf{r}) r^p \psi_{nlm}(\mathbf{r}) \, d^3 r$$

$$= \int_0^\infty \frac{P^*_{nl}(r)}{r} r^p \frac{P_{nl}(r)}{r} r^2 \, dr \underbrace{\int Y^*_{lm} Y_{lm} \, d\Omega}_{1} = \int_0^\infty |\, P_{nl}(r) \,|^2 \, r^p \, dr$$

From the explicit expression for $P_{nl}(r)$ and changing variables to $x = 2Zr/n$:

$$\langle r^p \rangle_{nl} = \underbrace{N^2_{nl} \left(\frac{n}{2Z} \right)^{p+1}}_{\propto Z \cdot Z^{-p-1} = Z^{-p}} \int_0^\infty x^{2l+2+p} \, e^{-x} \, F^2(-n+l+1, 2l+2, x) \, dx \quad \propto Z^{-p}$$

Known results:

$$\langle r \rangle_{nl} = \frac{1}{2Z} \left[3n^2 - l(l+1) \right] \qquad \langle r^2 \rangle_{nl} = \frac{n^2}{2Z^2} \left[5n^2 + 1 - 3l(l+1) \right]$$

$$\left\langle \frac{1}{r} \right\rangle_{nl} = \frac{Z}{n^2} \qquad \left\langle \frac{1}{r^2} \right\rangle_{nl} = \frac{Z^2}{n^3 (l + 1/2)}$$

Exercises

How to see dependence on nuclear charge Z from the Schrödinger equation?

Hydrogen: $\left(-\frac{\hbar^2 \nabla^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} \right)\psi = \varepsilon\psi$ Hydrogen-like: $\left(-\frac{\hbar^2 \nabla^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r} \right)\psi = \varepsilon\psi$

Introduce the scaled coordinate $\rho = Zr \quad \Rightarrow \quad \nabla_r^2 = Z^2 \nabla_\rho^2$

Hydrogen-like Schrödinger eq. becomes $Z^2\left(-\frac{\hbar^2 \nabla_\rho^2}{2m} - \frac{e^2}{4\pi\epsilon_0 \rho} \right)\psi = \varepsilon\psi$

Same solution as H but rescaled!

$\Rightarrow \quad E \propto Z^2$

Characteristic length scale for H atoms: Bohr radius $a_\mathrm{B} = 0.529 \cdot 10^{-10} m$

for H-like ions: a_B/Z

$\Rightarrow \quad \langle r^p \rangle \propto Z^{-p}$

Relativistic corrections

Velocity-dependence of mass:

The relativistic energy is given by $E^2 = p^2c^2 + m^2c^4$, and therefore the relativistic kinetic energy (subtracting the rest mass) is

$$K = \left(p^2c^2 + m^2c^4\right)^{1/2} - mc^2$$

$$= mc^2\left(1 + p^2/(m^2c^2)\right)^{1/2} - mc^2$$

$$\approx mc^2\left(1 + \frac{p^2}{2m^2c^2} - \frac{p^4}{8(m^2c^2)^2}\right) - mc^2$$

$$= \frac{p^2}{2m} - \frac{p^4}{8m^3c^2}$$

$$\left[(1+x)^{1/2} \simeq 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots\right]$$

$\Rightarrow$ relativistic correction $\Delta K = -\frac{1}{8}\frac{p^4}{m^3c^2}$

$\Rightarrow$ may introduce the perturbation $\delta V_m = -\frac{1}{8}\frac{p^4}{m^3c^2}$.

We can calculate this value from manipulation of the Schrödinger equation!

Relativistic corrections

$$\left(\frac{p^2}{2m} + V(r)\right)\psi_0 = \epsilon_0\psi_0$$

$$\Rightarrow \;\; p^2 = -\,2m\left(V(r) - \epsilon_0\right)$$

$$\Rightarrow \delta V_m = -\frac{1}{8}\frac{4m^2}{m^3c^2}\left(V(r)^2 - 2\epsilon_0 V(r) + \epsilon_0^2\right)$$

$$\Rightarrow \delta V_m = -\frac{1}{2}\frac{1}{mc^2}\left(V(r)^2 - 2\epsilon_0 V(r) + \epsilon_0^2\right)$$

Coulomb potential energy $V(r) = -\,Z/r$,

$\Rightarrow$ relativistic correction to the energy, $\Delta E_{\text{rel}} = \langle\psi_0\,|\,\delta V_m(r)\,|\,\psi_0\rangle$

$$\Rightarrow \;\; \Delta E_{\text{rel}} = -\frac{1}{2}\frac{Z^2}{mc^2}\left\langle \psi_0 \left| \frac{1}{r^2} \right| \psi_0 \right\rangle + \frac{1}{mc^2}\epsilon_0(-Z)\left\langle \psi_0 \left| \frac{1}{r} \right| \psi_0 \right\rangle - \frac{1}{2}\frac{1}{mc^2}\epsilon_0^2 \;\; .$$

Reminder: $\left\langle \frac{1}{r} \right\rangle_{nl} = \frac{Z}{n^2}$ and $\left\langle \frac{1}{r^2} \right\rangle_{nl} = \frac{Z^2}{n^3(l+1/2)}$

Relativistic corrections

Putting it together:

$$\Rightarrow \quad \Delta E_{\text{rel}} = -\frac{1}{2}\frac{1}{mc^2}\left(\frac{Z^4}{n^3(l+1/2)} + 2\epsilon_0\frac{Z^2}{n^2} + \epsilon_0^2\right)$$

$$\Rightarrow \quad \Delta E_{\text{rel}} = -\frac{1}{2}\frac{1}{mc^2}\left(\frac{Z^4}{n^3(l+1/2)} - 2\frac{Z^2}{2n^2}\frac{Z^2}{n^2} + \frac{Z^4}{4n^4}\right)$$

$$\Rightarrow \quad \Delta E_{\text{rel}} = -\frac{1}{2}\frac{1}{mc^2}\frac{Z^4}{n^3}\left(\frac{1}{l+1/2} - \frac{3}{4n}\right) = -\left(\frac{1}{l+1/2} - \frac{3}{4n}\right)\frac{Z^4\alpha^2}{n^3}\,\text{Ry}$$

$$= \left(\frac{1}{l+1/2} - \frac{3}{4n}\right)\frac{Z^2\alpha^2}{n}\,E_n$$

Where $\alpha = \frac{1}{4\pi\epsilon_0}\frac{e^2}{\hbar c} \approx 1/137$ is the fine-structure constant.

$\Delta E_{\text{rel}} \propto Z^4$ $\quad\Rightarrow\quad$ Relativistic effects are strongly enhanced in heavy ions.

Relativistic corrections

<u>Spin-orbit interaction:</u>

Electrons have spin $\Rightarrow$ magnetic moment $\boldsymbol{\mu} = -g_s \mu_B \mathbf{s}$,

where $\mu_B = e\hbar/(2mc)$ is the Bohr magneton, and $g_s \approx 2$ is the electron spin g-factor.

The electron magnetic moment $\boldsymbol{\mu}$ interacts with the magnetic field originating from the nucleus, leading to an interaction term $-\boldsymbol{\mu} \cdot \mathbf{B}$.

What is the source of the magnetic field? Because the electrons are moving at relativistic speeds, they see a B-field associated with the nuclear Coulomb (electric) field:

$$\mathbf{B} = -\frac{1}{c^2}\mathbf{v} \times \mathbf{E} \quad , \qquad \text{with} \qquad \mathbf{E} = \nabla V(r) = \frac{\partial V(r)}{\partial r}\mathbf{n} \qquad \text{(spherical symmetry)}$$

$$\Rightarrow \mathbf{B} = -\frac{1}{c^2}\mathbf{v} \times \left(\frac{\partial V(r)}{\partial r}\mathbf{n}\right) \qquad \mathbf{n} = \mathbf{r}/r$$

$$= -\frac{1}{c^2}\frac{1}{r}\frac{\partial V(r)}{\partial r}(\mathbf{v} \times \mathbf{r}) \qquad \mathbf{p} = m\mathbf{v}$$

$$= -\frac{1}{mc^2}\frac{1}{r}\frac{\partial V(r)}{\partial r}(\mathbf{p} \times \mathbf{r})$$

Relativistic corrections

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} = -\,\mathbf{p} \times \mathbf{r} \Rightarrow \qquad \mathbf{B} = -\frac{1}{mc^2}\frac{1}{r}\frac{\partial V(r)}{\partial r}(\mathbf{p} \times \mathbf{r}) = \frac{\hbar}{mc^2}\frac{1}{r}\frac{\partial V(r)}{\partial r}\mathbf{l}$$

Therefore, the interaction term: $V_{so} = -\boldsymbol{\mu} \cdot \mathbf{B} = \frac{e\hbar^2}{2m^2c^3}\frac{1}{r}\frac{\partial V(r)}{\partial r}\mathbf{s} \cdot \mathbf{l}$,

Note: a factor 1/2 has been inserted into the term above, due to accounting for Thomas precession.

This is a second relativistic correction to the gross energy structure, $\langle V_{so} \rangle \propto \left\langle \frac{1}{r}\frac{\partial V(r)}{\partial r}\mathbf{s} \cdot \mathbf{l} \right\rangle$.

$\Rightarrow$ $\mathbf{s}$ and $\mathbf{l}$ are coupled!

$\Rightarrow$ $\mathbf{j}$ is now a good quantum number, not $\mathbf{s}$ or $\mathbf{l}$.

What does the spin-orbit splitting look like?

$$\langle V_{so} \rangle \propto \left\langle \frac{1}{r}\frac{\partial V(r)}{\partial r} \right\rangle \langle \mathbf{s} \cdot \mathbf{l} \rangle \, .$$

Consider again point-nucleus Coulomb potential energy, $V_{\text{nuc}} = -\frac{Z}{r}$

$$\Rightarrow \langle V_{so} \rangle \propto Z \left\langle \frac{1}{r^3} \right\rangle_{nl} \langle \mathbf{s} \cdot \mathbf{l} \rangle \propto Z^4 \, .$$

What is $\langle \mathbf{s} \cdot \mathbf{l} \rangle$?

Relativistic corrections

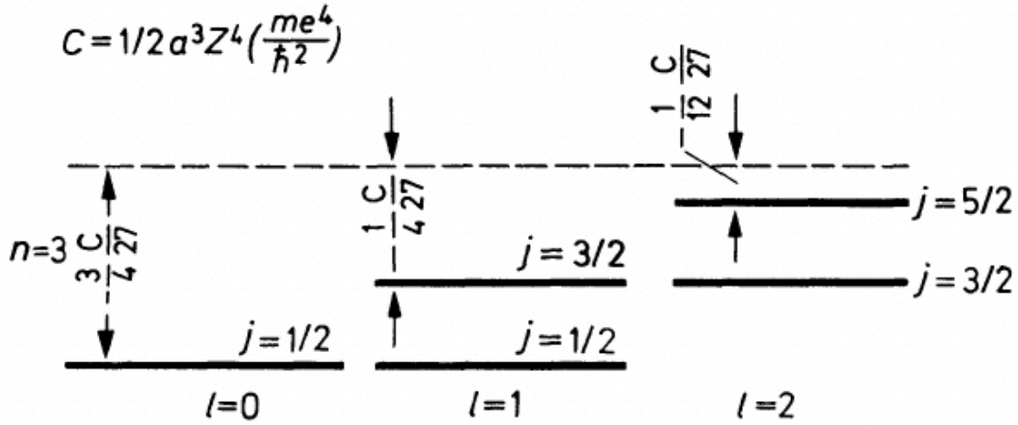
Exercise: calculate $\langle \mathbf{s}\cdot\mathbf{l}\rangle$.

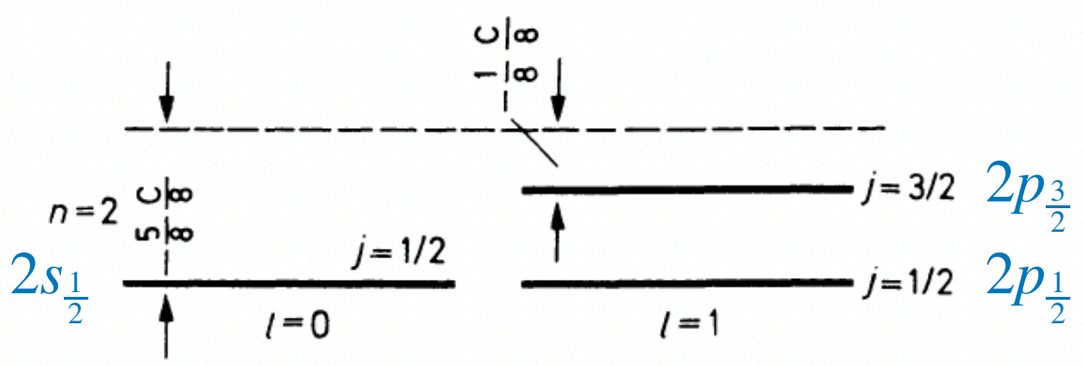
$$\mathbf{j} = \mathbf{l} + \mathbf{s}$$
$$\Rightarrow\ \mathbf{j}^2 = \mathbf{l}^2 + \mathbf{s}^2 + 2\mathbf{l}\cdot\mathbf{s}$$
$$\Rightarrow\ \mathbf{l}\cdot\mathbf{s} = \frac{1}{2}\left(\mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2\right)$$
$$\langle \mathbf{s}\cdot\mathbf{l}\rangle = \frac{1}{2}\left\langle \mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2\right\rangle$$
$$= \frac{1}{2}\left[j(j+1) - l(l+1) - s(s+1)\right]$$

For $s = 1/2$: $\quad j = l + \frac{1}{2} \ \Rightarrow\ \langle \mathbf{s}\cdot\mathbf{l}\rangle = \frac{l}{2}$

$$j = l - \frac{1}{2} \ \Rightarrow\ \langle \mathbf{s}\cdot\mathbf{l}\rangle = -\frac{l+1}{2}$$

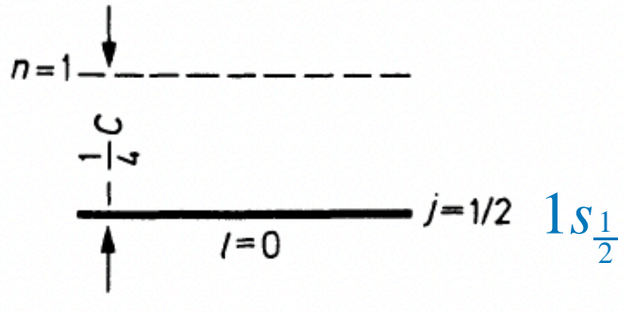
Fine structure





Total shift:

$$\Delta E_{\text{rel}} + \Delta E_{so} = -\left(\frac{1}{j+1/2} - \frac{3}{4n}\right)\frac{Z^4\alpha^2}{n^3}\,\text{Ry} = \left(\frac{1}{j+1/2} - \frac{3}{4n}\right)\frac{Z^2\alpha^2}{n}\,E_n$$



$\Rightarrow$ Binding energy: $\quad E_{nj} = \left[1 + \frac{Z^2\alpha^2}{n}\left(\frac{1}{j+1/2} - \frac{3}{4n}\right)\right]E_n$

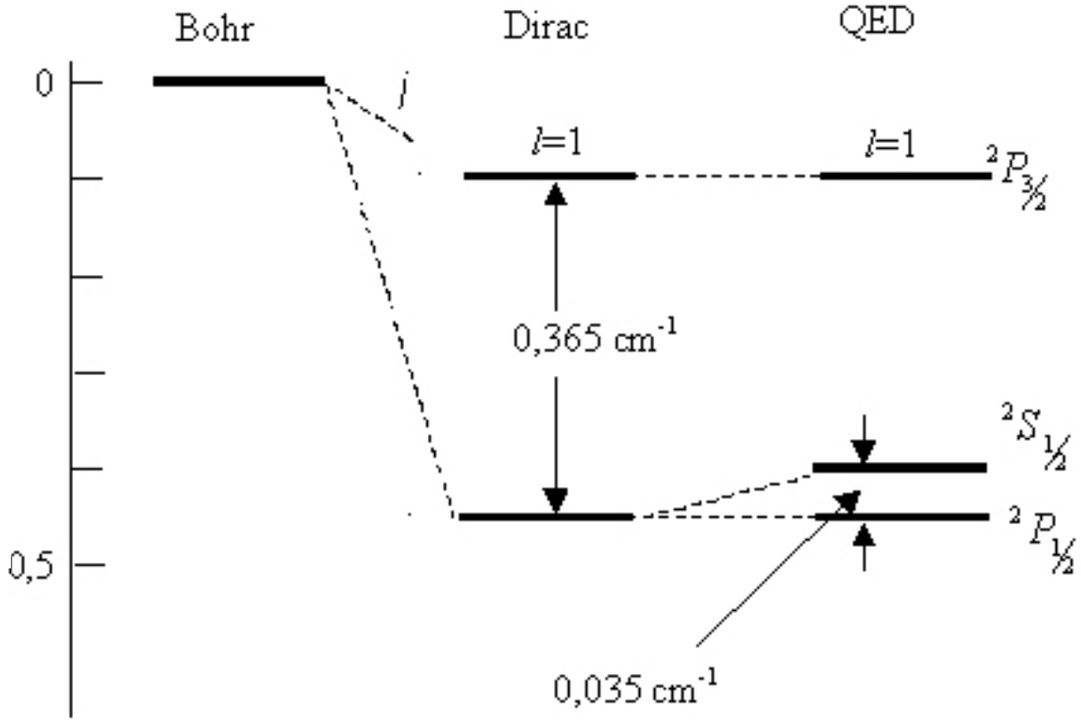
Lamb shift

Obtained expression for binding energy gives the same energy for states with the same n and j , though different l. E.g., for states $2s_{1/2}$ and $2p_{1/2}$.

In 1947, Lamb and Retherford measured a small splitting between $2s_{1/2}$ and $2p_{1/2}$ levels in H.

It could not be explained by any theory at the time.

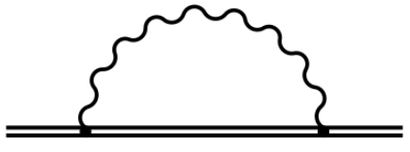
This measurement led to the development of quantum electrodynamics (QED) — the quantum theory of light and matter.



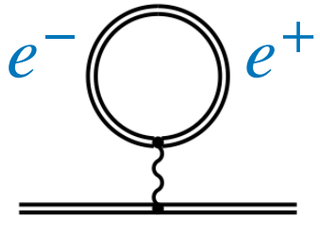
The Lamb shift is comprised of two types of QED radiative corrections: the *self-energy* and *vacuum polarisation*.

Lowest-order Feynman diagrams:

photon



Self-energy

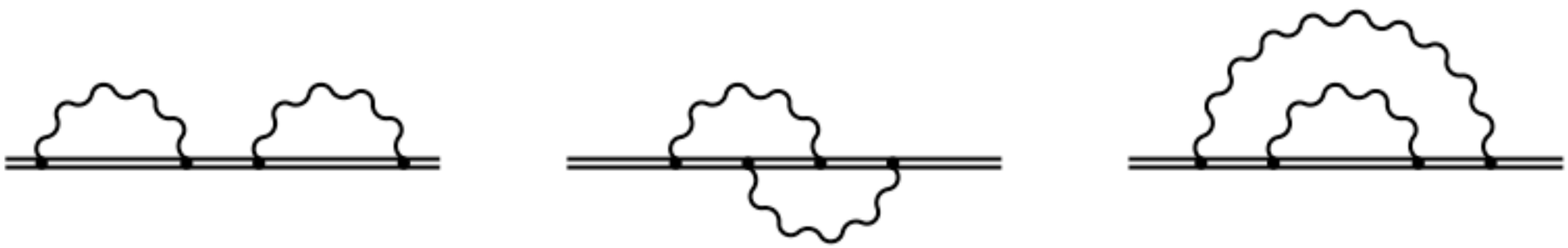


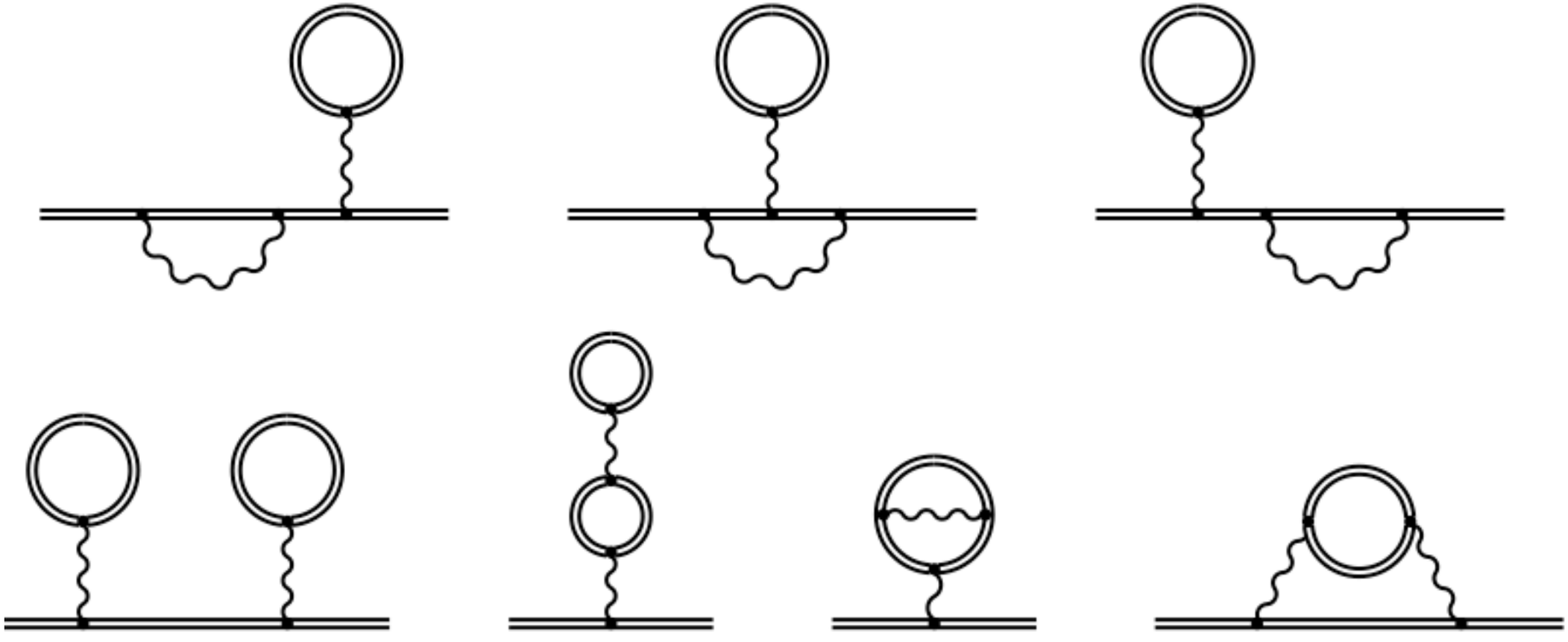
Vacuum polarisation

e^- in Coulomb field of nucleus

Lamb shift

Exercise: using the basic rules for constructing QED diagrams (discussed in class), draw the second-order QED radiative corrections to the energies.





Hyperfine structure

In atoms with non-zero total electronic angular momentum ***J*** and nuclei with non-zero spin ***I*** , the angular momenta couple to form total atomic angular momentum, $\mathbf{F} = \mathbf{I} + \mathbf{J}$.

$$|\mathbf{I} - \mathbf{J}| \leq \mathbf{F} \leq \mathbf{I} + \mathbf{J}$$

Similarly to the spin-orbit interaction, the interaction of ***I*** and ***J*** leads to a further splitting between energy levels — the hyperfine structure.

The interaction of the magnetic moment of the electrons with the magnetic field produced by the nuclear magnetic dipole moment may be quantified by the hyperfine constant A ,

$$\Delta E_{\mathrm{hf}} = A\langle \mathbf{I} \cdot \mathbf{J} \rangle$$

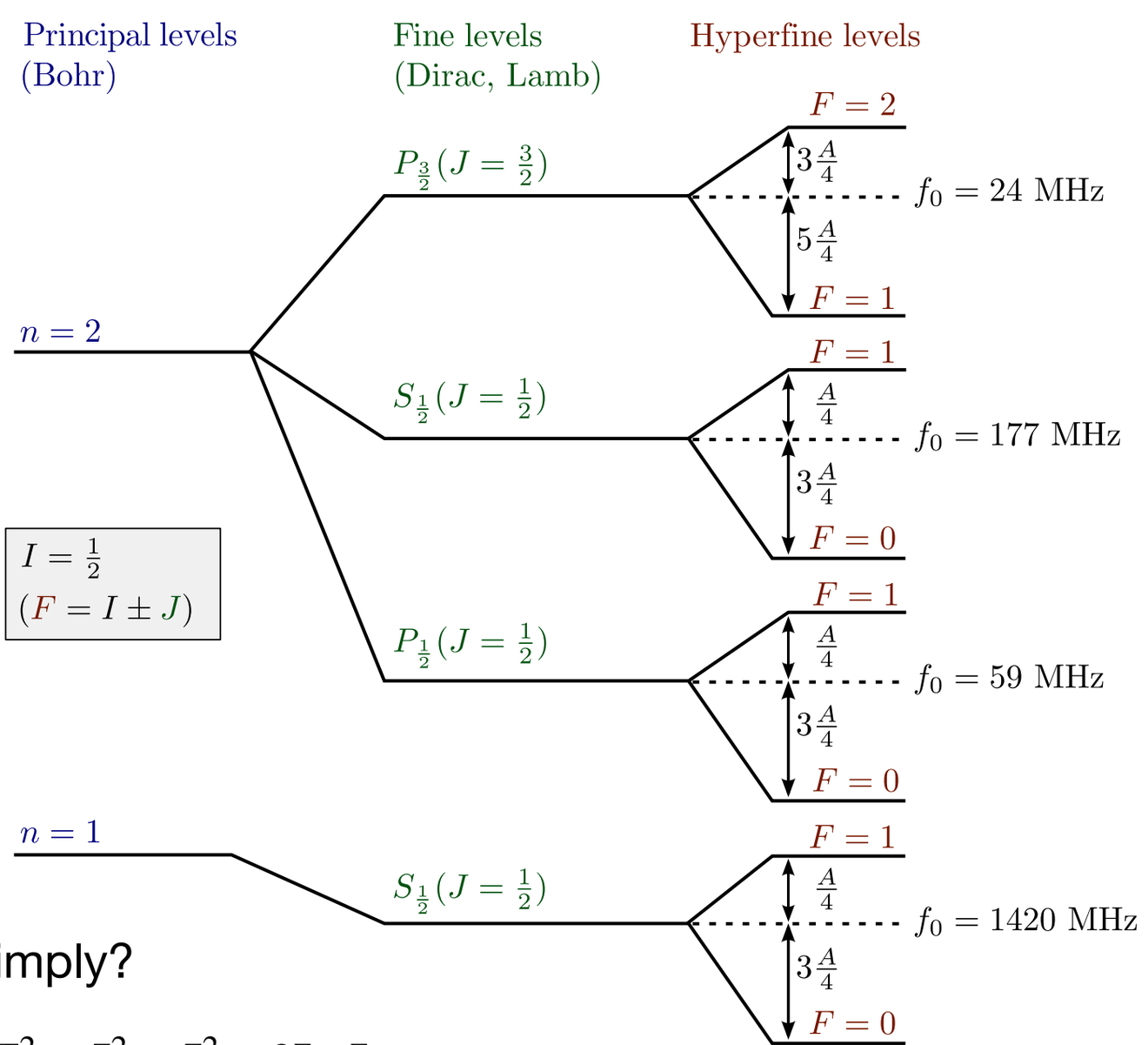


Image credit: Wiki Commons

Discussion: how can it be expressed so simply?

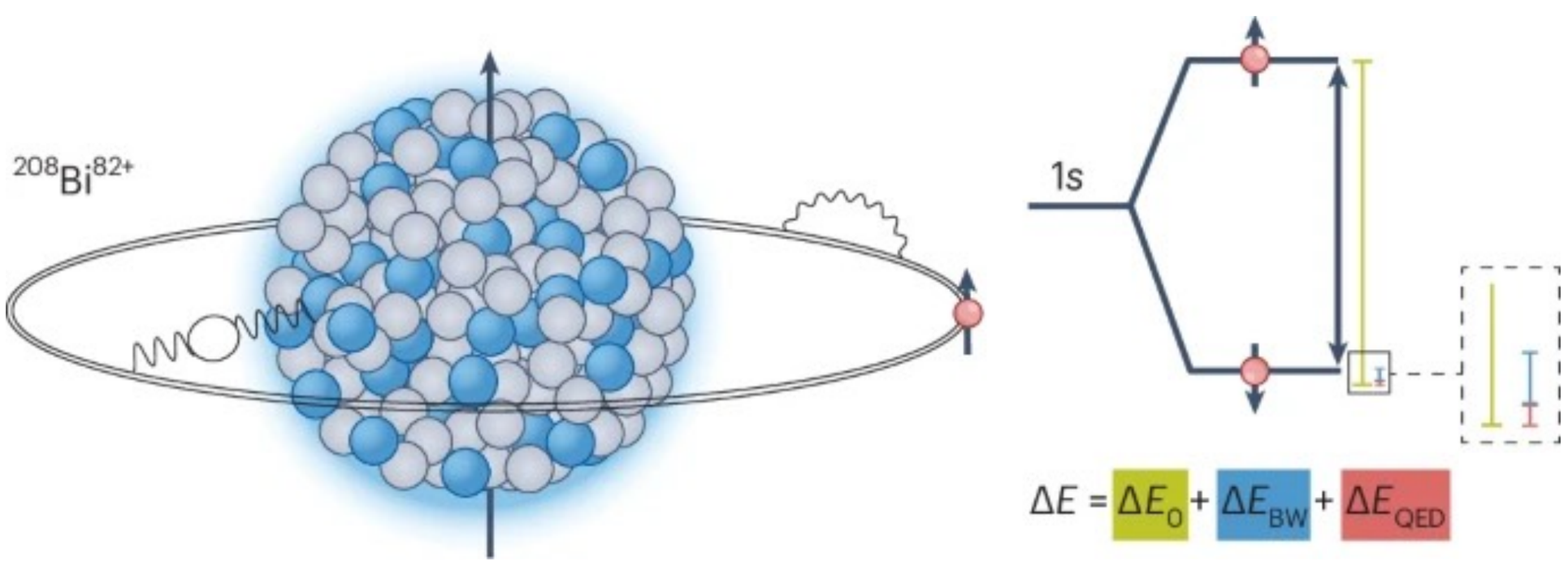
Exercise: calculate $\langle \mathbf{I} \cdot \mathbf{J} \rangle$. $\quad \mathbf{F} = \mathbf{I} + \mathbf{J} \;\Rightarrow\; \mathbf{F}^2 = \mathbf{I}^2 + \mathbf{J}^2 + 2\mathbf{I} \cdot \mathbf{J}$

$$\Rightarrow\; \mathbf{I} \cdot \mathbf{J} = \frac{1}{2}\left(\mathbf{F}^2 - \mathbf{I}^2 - \mathbf{J}^2\right) \;\Rightarrow\; \langle \mathbf{I} \cdot \mathbf{J} \rangle = \frac{1}{2}\left[F(F+1) - I(I+1) - J(J+1)\right]$$

QED tests

Find out more on QED tests in highly-charged (including H-like) ions? See, e.g., the review by Andrey Volotka *et al*., https://doi.org/10.1002/andp.201300079 .

For some more up-to-date works, and an accessible *News and Views* Jacinda wrote on a recent measurement of the hyperfine splitting in H-like bismuth-208, see https://www.nature.com/articles/s41567-025-02959-w .



By the end of this week, you should have an understanding of:

- H and H-like spectra
- Relativistic effects
- Fine structure
- Hyperfine structure

and be able to perform calculations related to these